

ALI ASAD HASSAN

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer with over 9 years of experience specializing in .NET Core, ASP.NET MVC, Web API, SQL Server, Oracle, Azure (Databricks, Data Factory), and Snowflake. Expertise in designing and optimizing enterprise-level applications, focusing on data integration, EDI (846/850/856/940) file processing, data migration, workflow automation, and cloud-based solutions. Proficient in supporting production-critical systems, monitoring scheduled jobs, and resolving complex issues within high-availability environments. Demonstrated leadership in mentoring developers and delivering scalable, secure, and reliable enterprise solutions.

WORK HISTORY

Senior Software Engineer, 02/2024 - Current

Algomus (Algo) – Lahore, Pakistan

Key Project : OneAlgo

- **Cloud Data Migration (Redshift to Databricks):** Successfully orchestrated the migration of legacy data architectures from AWS Redshift to Azure Databricks, streamlining big data processing using Apache Spark and Delta Lake.
- **Large-Scale Data Migration (Databricks to Snowflake):** Migrated over **1TB+ of enterprise data** from Azure Databricks to Snowflake, restructuring data models and optimizing query performance to deliver a **30% improvement** in downstream reporting speed.
- Built and optimized large-scale data pipelines using Azure Data Factory, Snowflake, SQL, and EDI integrations.
- Supported production-critical file-based and EDI integrations, performing root-cause analysis and resolving data pipeline issues across distributed systems.
- Monitored jobs, file flows, and system health using ActiveXperts while supporting secure AS2/EDI B2B integrations.
- Designed and implemented secure REST APIs using JWT Authentication, Refresh Tokens, and modern .NET technologies, enhancing application security and session management.
- Designed and developed Microservices using .NET, DDD, Clean Architecture, RabbitMQ, Redis, and Docker.

Senior Software Engineer, 01/2017 - 01/2024

Magnatec Systems – Lahore, Punjab, Pakistan

Key Project: Led the development of SMACC, 3SPOS and SCRIBES ERP solutions for Arab Sea Information Systems (Riyadh, KSA).

- Architected and developed a complete HRMS and ERP solution, managing end-to-end employee lifecycles, complex automated payroll engines, and cross-module business operations.
- Designed and developed complex SQL queries from scratch to generate critical financial accounting, real-time inventory management, and deep analytical business workflows.
- Optimized high-volume SQL Server databases through query tuning, advanced stored procedures, and triggers, significantly improving backend report generation speeds.
- Built scalable enterprise applications using ASP.NET WebForms/MVC, JavaScript, jQuery, and Bootstrap, ensuring seamless cross-browser layout stability.

- Led the full software development lifecycle (SDLC), including REST API development, third-party ERP integrations, stakeholder coordination, and developer mentoring.

Backend Development:

C#, ASP.NET Core, ASP.NET MVC, Web API, ADO.NET, LINQ, Entity Framework, RESTful API Development & Integration, Middleware Development & Request Pipeline Customization, FluentValidation, Serilog (Structured Logging), Exception Handling & Global Error Management

Frontend Technologies:

JavaScript, jQuery, AJAX, Angular 16/17/21, HTML, CSS, Bootstrap

Architecture & Design Patterns:

Microservices, Domain-Driven Design (DDD), Clean Architecture, Event-Driven Architecture, Unit of Work Pattern, Outbox Pattern, Singleton Pattern using double locking, Repository Pattern

Security & Authentication:

JWT Authentication, Refresh Token Implementation, Bcrypt (Password Hashing), Azure Key Vault (AKV), Rate Limiting, Role Based Access Control (RBAC), CORS, CSRF

Messaging & Caching:

RabbitMQ, Redis, Caching Strategies (L1/L2/L3)

Database Technologies:

SQL Server (Stored Procedures, MSDTC, Query Optimization, Linked Server, Triggers), Oracle Database, Snowflake (SaaS)

Cloud & Data Engineering:

Azure Databricks, Azure Data Factory (ADF) for ETL Pipelines, Azure Data Lake, Enterprise Integration & EDI (846/850/856/940 File Processing), Azure Blob Storage, Multi-layered Data Architectures (Landing, Processing, Archive), Access Control (IAM)

DevOps & Containerization:

Docker, GitHub, CI/CD Pipelines, SonarQube

Monitoring & Production Support:

Tidal Job Scheduling & Monitoring, ActiveXperts Monitoring & Alerting, Job/File/Service Health Monitoring, Production Logging, Root Cause Analysis & Troubleshooting

EDUCATION

BS(CS): Computer Science, 2014
University of South Asia – Lahore

KEY PROJECTS

- OneAlgo – https://one.algo.com/_admin/Login
- Scribes ERP – <https://scribeserp.com/en/>
- SMACC ERP – <https://www.smacc.com/en/>
- SMACC LAUNCHER - <https://configuration.smacc.com/>
- 3S POS – <https://3spos.com/en/>